EDA PROJECT

Udemy Courses

Exploratory Data Analysis

Uncovering insights in online learning trends, pricing strategies, and course performance metrics

3,678	4	13	2012–2017
Courses Analyzed	Subjects Covered	Data Columns	Time Period

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PROJECT OVERVIEW

What is this project about?

🎯 Objective

Analyze Udemy courses to uncover insights on performance, user preferences, and market trends in e-learning.

📊 Dataset

3,678 courses across 4 subjects with 13 attributes including price, subscribers, reviews & content duration.

🔍 Approach

End-to-end EDA: data understanding → cleaning → univariate → bivariate → temporal → correlation analysis.

💡 Goal

Extract actionable insights to help instructors & platforms optimize course offerings and strategy.

DATASET DESCRIPTION

Column	Type	Description
course_id	int64	Unique identifier for each course
course_title	object	Title of the course
is_paid	bool	Indicates paid or free course
price	int64	Price of the course in USD
num_subscribers	int64	Total number of students enrolled
num_reviews	int64	Number of reviews received
num_lectures	int64	Total number of lectures
course_level	category	Difficulty level (Beginner / Intermediate / Expert)
content_duration	float64	Duration of course content in hours
Date	datetime64	Date when course was published

3,678 • Total Records

13 • Columns

0 • Null Values

DATA CLEANING

STEP 01

Null Value Check

Verified all 13 columns — zero null values found across all 3,678 rows. Dataset is complete and requires no imputation.

✅ No Missing Values

STEP 02

Data Type Conversion

Converted 'Date' column from object → datetime64 for temporal analysis. Converted 'course_level' from object → category for memory efficiency.

🔚 2 Columns Converted

STEP 03

Redundant Column Removal

Dropped 3 non-analytical columns: course_id, course_title, and url. These don't contribute to statistical analysis.

🗑️ 3 Columns Dropped

STEP 04

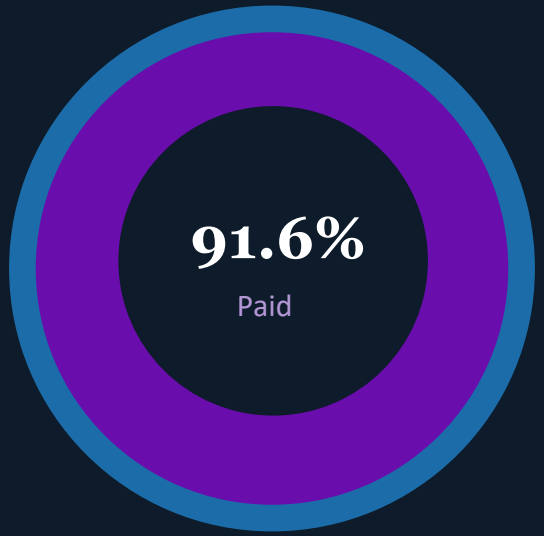
Clean Dataset

Final dataset: 3,678 rows × 10 columns, with proper datatypes and zero missing data — ready for analysis.

🎯 10 Columns Retained

UNIVARIATE ANALYSIS

Free vs Paid Courses



91.6%

Paid Courses

~3,372 courses

8.4%

Free Courses

~306 courses

Key Insight: Paid courses dominate the Udemy platform, making up over 90% of all available content. Free courses are rare and primarily used as lead-generation tools.

SUBJECT ANALYSIS — Subscribers & Revenue

Web Development

Subscribers 7.8M

Revenue $295K

Business Finance

Subscribers 2.9M

Revenue $280K

Musical Instruments

Subscribers 2.1M

Revenue $105K

Graphic Design

Subscribers 1.6M

Revenue $130K

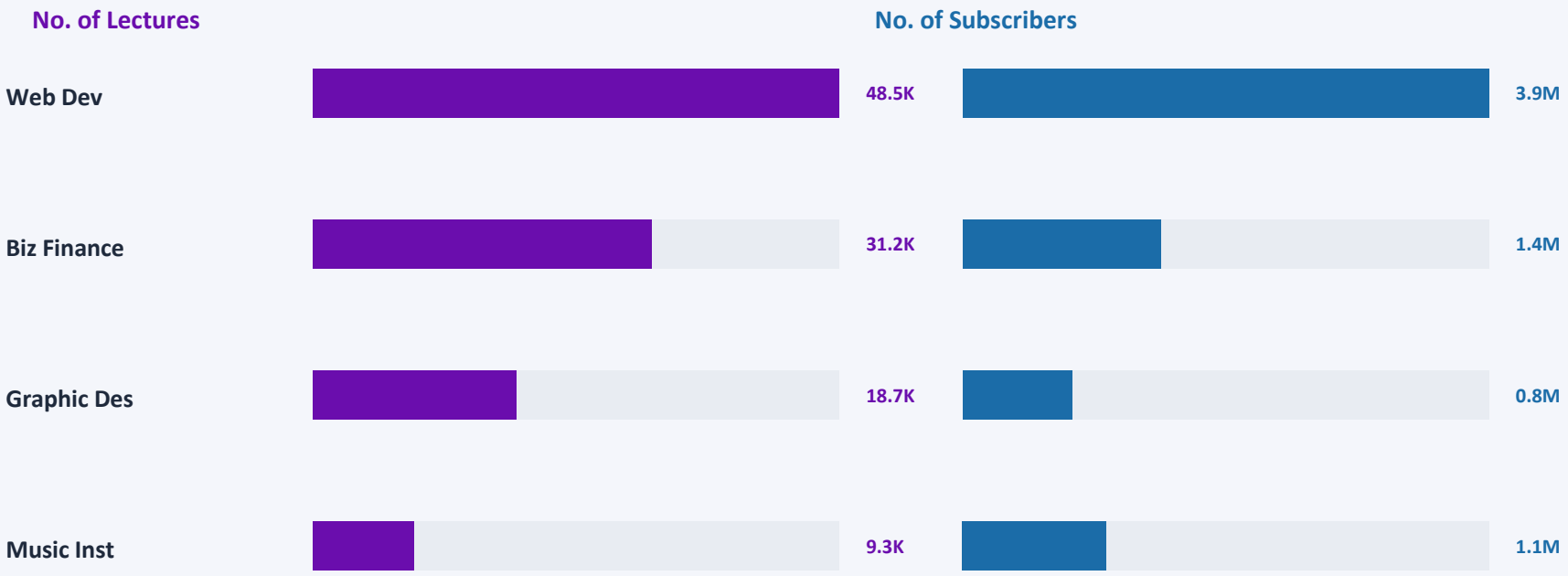
★ ***Web Development leads in both subscribers and revenue, while Business Finance ranks 2nd in both metrics.***

COURSE LEVEL ANALYSIS

Distribution by Difficulty Level

💡 "All Levels" is the most common category (53%), reflecting instructors' desire to maximize reach. Beginner content is the next biggest draw at 32%.

BIVARIATE ANALYSIS — Lectures & Subscribers by Subject + Level



Key Findings:

① Web Development dominates both lecture count & subscriber numbers.

② Musical Instruments has fewer lectures but relatively high subscribers — niche but popular.

③ Business Finance: high courses & revenue but fewer total subscribers than Web Dev.

CORRELATION ANALYSIS — Numeric Feature Relationships

	Price	Subscribers	Reviews	Lectures	Duration	Year
Price	**1.00**	0.01	0.02	0.10	0.09	-0.05
Subscribers	0.01	**1.00**	0.86	0.11	0.12	0.02
Reviews	0.02	0.86	**1.00**	0.09	0.10	0.04
Lectures	0.10	0.11	0.09	**1.00**	0.64	0.07
Duration	0.09	0.12	0.10	0.64	**1.00**	0.01
Year	-0.05	0.02	0.04	0.07	0.01	**1.00**

Key Correlations:

Subscribers ↔ Reviews — r = **0.86**

Very strong positive link

Lectures ↔ Duration — r = **0.64**

Longer courses = more lectures

Price ↔ Subscribers — r = **0.01**

No meaningful correlation

RECOMMENDATIONS

Actionable Insights for Instructors & Platforms

🎯 01

Target High-Demand Subjects

Focus on Web Development & Business Finance — they drive the highest engagement and revenue on the platform.

💰 02

Hybrid Pricing Strategy

Offer free introductory courses to attract learners and paid advanced modules to convert and monetize them.

🌱 03

Prioritize Beginner Content

Beginner-level courses attract the largest audience (32%). Design content that is accessible and welcoming.

⭐ 04

Leverage User Reviews

Reviews have a strong correlation (r=0.86) with subscribers. Encourage early learners to leave feedback.

📋 05

Optimize Course Structure

Balance lecture count and duration. Longer, well-structured courses correlate with better engagement.

📊 06

Use Data-Driven Decisions

Monitor year-over-year trends and adjust strategy based on subscriber and pricing publishing performance.

CONCLUSION

> This EDA highlights key factors driving success on Udemy — including subject demand, pricing strategy, course structure, and user engagement. Paid courses dominate the platform, while beginner-level content attracts the majority of learners. Strong correlations between subscribers and reviews emphasize the importance of quality and feedback. A data-driven approach is essential for optimizing course performance and enhancing learner experience.

5 Key Takeaways

- Paid courses dominate (91.6%) the platform
- Web Development leads in subs & revenue
- Peak activity was in 2015–2016
- Reviews strongly drive subscriber growth
- Beginner content attracts the widest audience

THANK YOU

Udemy Courses EDA Project

Key Topics Covered:

Data Cleaning • Univariate Analysis • Bivariate Analysis

Temporal Trends • Correlation Analysis • Recommendations

Built with Python • Pandas • Matplotlib • Seaborn